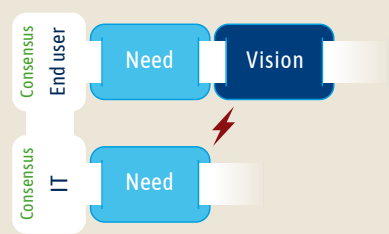


THE ALIGNMENT PUZZLE

WHITE PAPER

Why so many IT projects fail



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1. Introduction

Implementing new software is a change process. We prefer such processes to be controlled and successful. But IT projects have a bad reputation. As one of the authors for *The Alignment Puzzle* wrote: “Every IT project larger than π (3.14) man years tends to be delayed by a factor of π .”

Governments, in particular, regularly make the news. IT projects there seem to fail one after the other.

A few examples, just for the Dutch government:

- The Justice Department’s digital litigation system (planned 8 million euros, actual 29 million)
- The INSPECT project of the Netherlands Food and Consumer Product Safety Authority (budgeted 36 million euros, actual 95 million, without usable result)
- The tax allowance system of the Ministry of Finance (planned 60 million euros, actual 240 million)
- The SPEER project for Defence (planned 200 million euros, actual 1 billion¹)

The government is apparently an environment where projects easily slip out of management’s hands. Projects are too big and take too long. Requirements and wishes are adjusted along the way under pressure from all kinds of stakeholders. There is a lack of knowledge, measurable objectives, and a clear long-term vision.

But it is not only in the government that many IT projects fail. We cite the failed SAP implementation at the German supermarket chain Lidl as an example. At first glance, that major project looked like a good match between two top German companies, but it was ultimately called off in 2018 after half a billion euros (a five with eight zeros, for clarity) had been spent.

Excessive lead times and too many involved parties are undoubtedly two of the main reasons for failure. But there is another very important reason: failing to meet the five basic ingredients for successful change. To illustrate this, we use the ‘Knoster model’².

1 Note that the factor π we mentioned earlier is a modest and conservative guess compared to the results in this list.

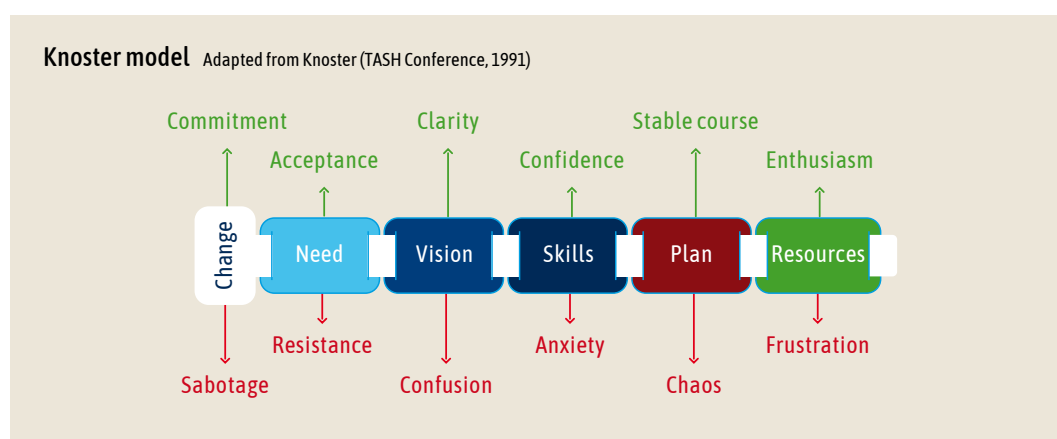
2 T. Knoster (June 1991). Factors in managing complex change. Material presentation at TASH conference, Washington D.C., The Association for People with Severe Disabilities.

2. The Knoster Model

This model describes five basic ingredients needed to bring about successful change: need, vision, skills, plan, and resources. The model describes the positive effects when the ingredients are present in the right quantity and form, but also the negative effects that occur when one or more of the ingredients are missing. And those are a threat to the project.

- If the **need** or necessity is not clear, people conclude that the new situation might be a setback, and that can evoke resistance.
- If the **vision** is lacking, it is unclear to people where things are heading and what is going to happen. This creates confusion.
- If people think they do not possess the necessary **skills** to act in the new environment, anxiety arises.
- Without a solid **plan**, people may assume that it will result in chaos.
- And if people notice that there are no or insufficient **resources** to get things done, frustration will prevail.

If one or more of the five principles are unclear, this can lead to people resisting or even outright sabotaging the change.



The interesting thing about this analysis is that it goes a step further than merely noting that people have something like an ‘inherent resistance to change’. There are various possible negative reactions, each with a cause, which therefore need to be addressed.

You can recognize confusion, anxiety, chaos, and frustration when talking to people. If you notice these situations within the context of a change process, you can determine early in the process whether or not sufficient attention has been paid to one or more of the five principles mentioned.

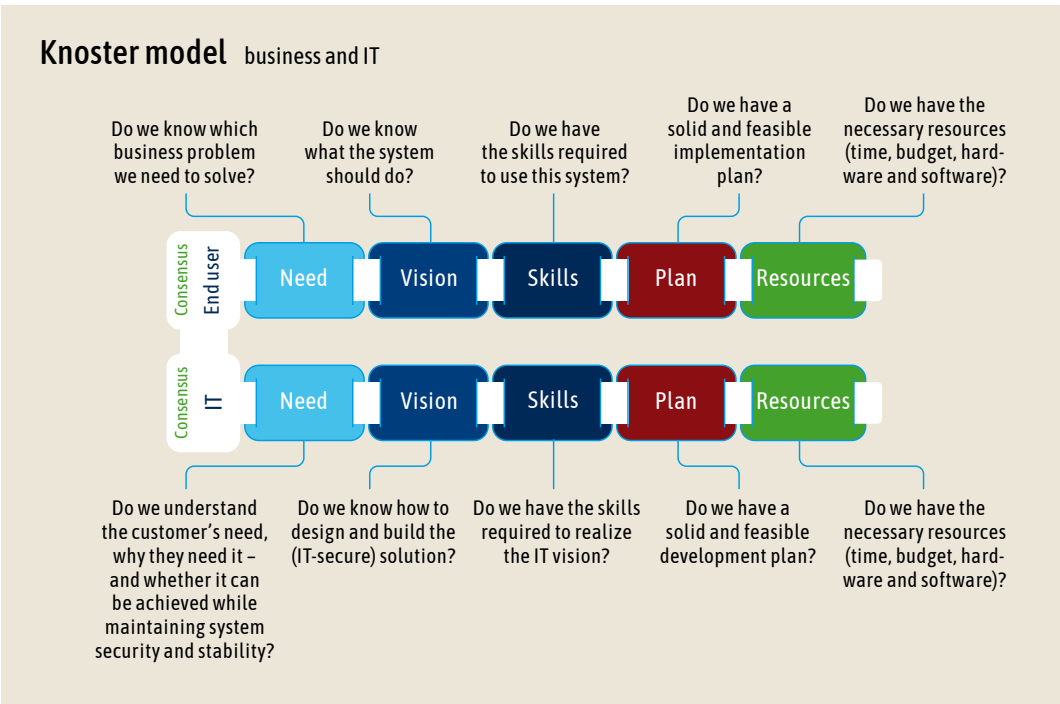
If it is merely a matter of perception, you take the necessary steps to adjust that perception. If it turns out that the principle being questioned has not been sufficiently developed and substantiated, then (re)defining and (re)communicating it is the next step.

3. The Knoster Model for IT Projects

The Knoster framework applies to changes that must be brought about by individuals. However, individuals are always part of stakeholder groups, such as sales, production, accounting, and IT. Suppose the change we want to bring about is ‘designing, building, and implementing a correctly functioning software system that fully meets the needs of the users’. It immediately becomes apparent that you are dealing with various stakeholder groups for this issue. The question then is how to apply Knoster to this.

In the simple case of an end user requesting new or modified software functionality, we have two stakeholder groups – the end user and one or more ‘IT professionals’. Each of the participating individuals must then have a clear understanding of the five principles for their own area of application. Additionally, the participants from the two areas of application must reach a consensus on exactly what needs to be delivered. You must therefore pose and answer the Needs-Vision-Skills-Plan-Resources questions for both the end user and IT.

Let’s take a look at ‘Business-IT misalignment’ through this lens in terms of Needs and Vision.

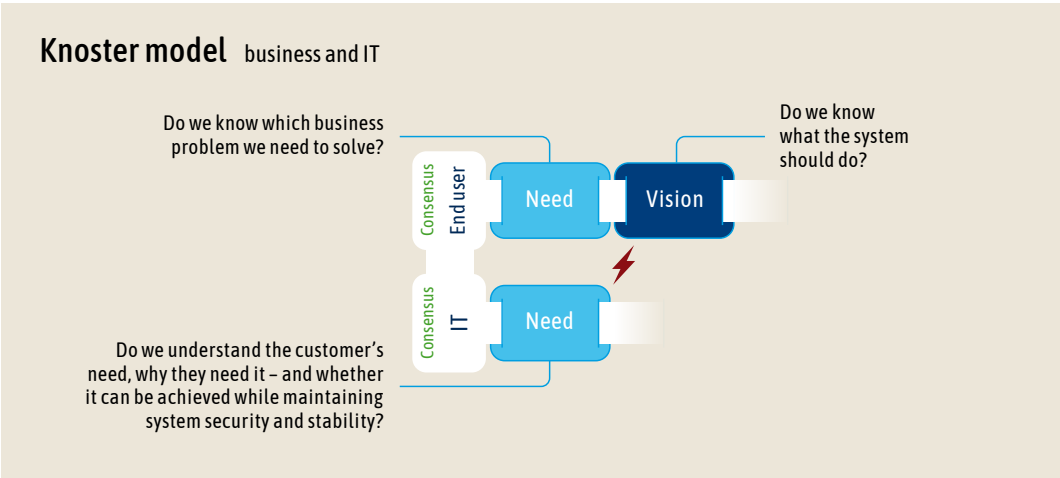


4. The needs and visions of Business and IT

Generally speaking, business users require new or modified functionality because they are convinced that this will help them with their daily work. Since the world and the environment are constantly changing, they prefer a flexible software system that can easily be adapted to new needs. And because that environment is constantly changing, they also expect fast response times from IT.

Naturally, IT wants to serve the business, but they too have to take IT-specific top priorities into account. In many companies, the performance of IT departments is initially judged on security, stability, and cost control. This is one of the main reasons why IT is generally averse to ‘new software’. Rather than jumping for joy because they get to make or buy that new software for you – which subsequently gets to be installed on yet another new server – IT professionals are more likely to look at your plans critically out of fear of security breaches or reduced stability.

We have previously argued that the mantra ‘The business specifies and IT realizes’ does not work³. In the Business-IT Knoster diagram, this problem is situated in the end user’s Vision block and the IT Needs block. In most cases, end users know which business problem they want to solve, which can range from simple things like performing transactions faster or with a lower risk of error, to more complex matters such as introducing a new support service for customers. The issue here lies more in whether they can clearly visualize what the system is actually supposed to do for them and how it should do it. For example, we have noticed that people fall back on techniques they know when asking questions, partly because they are relatively unfamiliar with new developments within IT.



3 The Alignment Puzzle (2025), chapter 13.3.2

If the end user does not sufficiently know or cannot articulate what the system should do (= determining what's in the Vision block), then it is difficult for IT to answer the question 'Do we know the customer's needs?' (= the first part of the IT Needs block). If Business and IT are not aware of this, it is not surprising that what is delivered does not match expectations.

One way to deal with this problem is the aforementioned Rapid Prototyping.

If IT has a clear picture of the end user's Needs and Vision, they can proceed with defining the IT Vision, assessing and potentially adjusting the necessary skills, the implementation plan, and making the necessary resources available.

As soon as IT has a working system (or at least a version that can be used to compile user training), Business can proceed with their part of the skills, plan, and resources.



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Design & production: Okapi Ontwerp & dtp.

Images: Illustrations and infographics by Fons Moers / Okapi.